

CSCI3230 (ESTR3108)

Fundamentals of Artificial Intelligence

Tutorial 1

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Outline

Part 1. Review linear algebra

Part 2. Least squares

Part 3. Bias-variance decomposition



Part 1. Review linear algebra

- Notation for vectors in $\mathbb{R}^n$:

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \quad \text{column vector}$$

$$X' = (x_1, \dots, x_n) \quad \text{row vector}$$

- Transpose: column vector $\leftrightarrow$ row vector

$$X' = X^T, \quad X'^T = X$$

- In the context of vectors, column vectors and row vectors are the same. But when putting them with matrices, they are totally different.

- Scaling: $a \in \mathbb{R}$, $X = (x_1, \dots, x_n) \in \mathbb{R}^n$, $aX = (ax_1, \dots, ax_n)$
- Addition: $X = (x_1, \dots, x_n)$, $Y = (y_1, \dots, y_n)$,
 $X + Y = (x_1 + y_1, \dots, x_n + y_n)$
- Suppose we have m vectors: $X_1 \dots X_m$, linear combination of these vectors: for any m scalars $a_1 \dots a_m$,

$$a_1 X_1 + \dots + a_m X_m$$

- Inner product (dot product): $\forall X \in \mathbb{R}^n, Y \in \mathbb{R}^n$

$$X \cdot Y = X^T Y = \sum_{i=1}^n x_i y_i$$

- Euclidean norm: $\|X\|_2 = \sqrt{X^T X} = \sqrt{\sum_{i=1}^n x_i^2}$

Matrix

- Notation for $m \times n$ matrices:

$$\mathbf{A} = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix}$$

$a_{i,j}$ is the element at the i -th row, j -th column of $\mathbf{A}$.

- Denote the i -th column of $\mathbf{A}$ as $\mathbf{A}_i$ (column vectors), and j -th row of $\mathbf{A}$ as $\mathbf{A}^{(j)}$ (row vectors).
- Transpose: columns $\leftrightarrow$ rows

$$\mathbf{A}^T = \begin{pmatrix} a_{1,1} & \cdots & a_{m,1} \\ \vdots & \ddots & \vdots \\ a_{1,n} & \cdots & a_{m,n} \end{pmatrix} = [A^{(1)T}, \dots, A^{(m)T}]$$

Matrix

- n-dimensional row vector: $1 \times n$ matrix
- n-dimensional column vector: $n \times 1$ matrix

- Scaling: $c \in \mathbb{R}$, $\mathbf{A} = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix}$, $c\mathbf{A} = \begin{pmatrix} ca_{1,1} & \cdots & ca_{1,n} \\ \vdots & \ddots & \vdots \\ ca_{m,1} & \cdots & ca_{m,n} \end{pmatrix}$

- Addition: $\mathbf{A} = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} b_{1,1} & \cdots & b_{1,n} \\ \vdots & \ddots & \vdots \\ b_{m,1} & \cdots & b_{m,n} \end{pmatrix}$

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} a_{1,1} + b_{1,1} & \cdots & a_{1,n} + b_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} + b_{m,1} & \cdots & a_{m,n} + b_{m,n} \end{pmatrix}$$

$\mathbf{A}$ and $\mathbf{B}$ have the same shape: $\mathbb{R}^{m \times n}$

Special matrices

- Diagonal matrix in $\mathbb{R}^{n \times n}$:

$$\Lambda = \begin{pmatrix} \Lambda_1 & 0 & \cdots & 0 \\ 0 & \Lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \Lambda_n \end{pmatrix}$$

- Identity matrix in $\mathbb{R}^{n \times n}$: $I = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$

- Symmetric matrix: $A = A^T$

Matrix-vector multiplication

- Matrix-vector multiplication

$$\begin{aligned}\mathbf{A}X &= \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\ &= \begin{pmatrix} a_{1,1} \\ \vdots \\ a_{m,1} \end{pmatrix} x_1 + \begin{pmatrix} a_{1,2} \\ \vdots \\ a_{m,2} \end{pmatrix} x_2 + \cdots + \begin{pmatrix} a_{1,n} \\ \vdots \\ a_{m,n} \end{pmatrix} x_n \\ &= \mathbf{A}_1 x_1 + \cdots + \mathbf{A}_n x_n\end{aligned}$$

- Notice the shape: $\mathbf{A} \in \mathbb{R}^{m \times n}$, $X \in \mathbb{R}^n$, $\mathbf{A}X \in \mathbb{R}^m$.
- Matrix-vector multiplication = linear combination of $\mathbf{A}$'s columns.

Matrix-vector multiplication

Another perspective of matrix-vector multiplication:

- Matrix-vector multiplication

$$\begin{aligned}\mathbf{A}X &= \begin{pmatrix} \mathbf{A}^{(1)} \\ \vdots \\ \mathbf{A}^{(m)} \end{pmatrix} X \\ &= \begin{pmatrix} \mathbf{A}^{(1)} \cdot X \\ \vdots \\ \mathbf{A}^{(m)} \cdot X \end{pmatrix}\end{aligned}$$

Recall that $\mathbf{A}^{(i)}$ is the i -th row of $\mathbf{A}$.

- Matrix-vector multiplication = dot product of $\mathbf{A}$'s rows and X .

Matrix-matrix multiplication

- Matrix-matrix multiplication

$$\mathbf{AX} = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix} \begin{pmatrix} x_{1,1} & \cdots & x_{1,l} \\ \vdots & \ddots & \vdots \\ x_{n,1} & \cdots & x_{n,l} \end{pmatrix}$$

- Let X_i be the i -th column of $\mathbf{X}$, the i -th column of $\mathbf{AX}$ is $\mathbf{AX}_i$:

$$\mathbf{AX} = [\mathbf{AX}_1, \dots, \mathbf{AX}_l]$$

- Notice the shape: $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{X} \in \mathbb{R}^{n \times l}$, $\mathbf{AX} \in \mathbb{R}^{m \times l}$.
- Multiplication for block matrices: $\mathbf{A} = [\mathbf{A}_1, \mathbf{A}_2]$

$$\mathbf{BA} = \mathbf{B}[\mathbf{A}_1, \mathbf{A}_2] = [\mathbf{BA}_1, \mathbf{BA}_2]$$

Inverse matrix

- For squared matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, if there is a $\mathbf{B} \in \mathbb{R}^{n \times n}$ such that

$$\mathbf{AB} = \mathbf{I} = \mathbf{BA}$$

$\mathbf{A}$ is invertible and $\mathbf{A}^{-1} = \mathbf{B}$

- How to find inverse matrix of $\mathbf{A}$ (if invertible):
 - 1 Gauss-Jordan elimination: find a row-operating matrix $\mathbf{B}$ which transforms $\mathbf{A}$ to $\mathbf{I}$. (More feasible for human beings)
 - 2 Use eigen-decomposition: $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$, $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Lambda}^{-1}\mathbf{U}^{-1}$
 - 3 Use the analytic solution:

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \mathbf{C}^T$$

where $\mathbf{C}$ is the cofactor matrix of $\mathbf{A}$.

Matrix calculus

- Univariate real-valued function $f(x) : \mathbb{R} \rightarrow \mathbb{R}$
- Multi-variate real-valued function $f(x_1, \dots, x_n) : \mathbb{R}^n \rightarrow \mathbb{R}$.
For convenience, we use $X = (x_1, \dots, x_n)^T$ to represent the independent variable. So we can write it as $f(X) : \mathbb{R}^n \rightarrow \mathbb{R}$.
- We use denominator layout to find the derivative w.r.t. X .

$$\frac{\partial f(X)}{\partial X} = \left(\frac{\partial f(X)}{\partial x_1}, \frac{\partial f(X)}{\partial x_2}, \dots, \frac{\partial f(X)}{\partial x_n} \right)^T$$

Due to denominator layout, the first derivative is a column vector.

Reference: [Wikipedia - Matrix calculus](#)

Matrix calculus

$\forall X \in \mathbb{R}^n, Y \in \mathbb{R}^n$:

$$\frac{\partial Y^T X}{\partial X} = Y$$

$$\frac{\partial X^T Y}{\partial X} = Y$$

$$\frac{\partial X^T \mathbf{A} X}{\partial X} = (\mathbf{A} + \mathbf{A}^T) X$$

Matrix calculus

- Derivative of a linear form

$$Y^T X = \sum_{i=1}^n y_i x_i$$

Hence, for each x_k ,

$$\frac{\partial(Y^T X)}{\partial x_k} = y_k.$$

Therefore,

$$\frac{\partial Y^T X}{\partial X} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = Y$$

Since $X^T Y$ is a scalar,

$$\frac{\partial X^T Y}{\partial X} = \frac{\partial (X^T Y)^T}{\partial X} = \frac{\partial Y^T X}{\partial X} = Y$$

- Derivative of a quadratic form

$$f(X) = X^T \mathbf{A} X = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j.$$

Consider the derivative with respect to one component x_k :

$$\begin{aligned} \frac{\partial f(X)}{\partial x_k} &= \frac{\partial}{\partial x_k} \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j = \sum_{j=1}^n a_{kj} x_j + \sum_{i=1}^n a_{ik} x_i \\ &= (\mathbf{A} X)_k + (\mathbf{A}^T X)_k. \end{aligned}$$

Since this holds for every $k = 1, \dots, n$,

$$\frac{\partial X^T \mathbf{A} X}{\partial X} = \mathbf{A} X + \mathbf{A}^T X = (\mathbf{A} + \mathbf{A}^T) X.$$

Linear algebra materials

- 1 Introduction to Linear Algebra. Gilbert Strang.
- 2 Matrix Analysis and Applied Linear Algebra. Carl D. Meyer.
- 3 Advanced Linear Algebra. Steven Roman.
- 4 **Linear Algebra and Its Applications.** Manolis C. Tsakiris.



Part 2. Least squares

Problem settings

- Recall that our data matrix $\mathbf{X}$ and observed labels Y :

$$\mathbf{X} = \begin{pmatrix} \mathbf{1} & x_1^{(1)} & \dots & x_n^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{1} & x_1^{(m)} & \dots & x_n^{(m)} \end{pmatrix} \quad Y = \begin{pmatrix} y^{(1)} \\ \vdots \\ y^{(m)} \end{pmatrix}$$

- We use linear function to fit a linear mapping from $\mathbf{X}$ to Y :

$$\hat{Y} = \mathbf{X}\Theta$$

where $\Theta = \begin{pmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_n \end{pmatrix}$, such that $\hat{Y}$ is very close to Y .

Ordinary least squares

- Recall that we use $\|\hat{Y} - Y\|_2^2$ to measure the distance between real labels and estimated labels.
- Note that:

$$\|\hat{Y} - Y\|_2^2 = \sum_{i=0}^m (\hat{y}^{(i)} - y^{(i)})^2$$

which is exactly the residual sum of squares (RSS).

- Ordinary least squares (OLS) estimator:

$$\hat{\Theta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T Y$$

is the optimal solution that minimizes $\|\hat{Y} - Y\|_2$.

Derivation

Analytic solution:

$$\begin{aligned} J(\Theta) &= \|\hat{f}_{\Theta}(\mathbf{X}) - Y\|_2^2 = (\mathbf{X}\Theta - Y)^T(\mathbf{X}\Theta - Y) \\ &= \Theta^T \mathbf{X}^T \mathbf{X} \Theta - Y^T \mathbf{X} \Theta - \Theta^T \mathbf{X}^T Y + Y^T Y \end{aligned}$$

$$\begin{aligned} \frac{\partial J(\Theta)}{\partial \Theta} &= 2\mathbf{X}^T \mathbf{X} \Theta - \mathbf{X}^T Y - \mathbf{X}^T Y \\ &= 2\mathbf{X}^T (\mathbf{X} \Theta - Y) = 0 \dots\dots\dots (1) \end{aligned}$$

$$\frac{\partial^2 J(\Theta)}{\partial \Theta^2} = 2\mathbf{X}^T \mathbf{X} \succeq 0 \text{ is for true.}$$

$$\text{By (1), } \mathbf{X}^T \mathbf{X} \Theta = \mathbf{X}^T Y \implies \boxed{\hat{\Theta} = \Theta^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T Y}$$

Example

We have 3 samples with bi-variate features:

$$X^{(1)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad X^{(2)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad X^{(3)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

with corresponding observed labels: $y^{(1)} = 1, y^{(2)} = 2, y^{(3)} = 3$. Suppose we use linear regression model to predict the relationship between the features and labels. Please use OLS to find the $\hat{\Theta}$.

Example (cont.)

- Recall that the linear regression model contains an intercept term:

$$\hat{Y} = \mathbf{X}\Theta, \quad \Theta = \begin{pmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{pmatrix}.$$

- According to the three samples, the design matrix and label vector are

$$\mathbf{X} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

- Using the OLS solution

$$\hat{\Theta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T Y,$$

we first calculate

$$\mathbf{X}^T \mathbf{X} = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{X}^T Y = \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix}.$$

Example (cont.)

- The inverse of $\mathbf{X}^T\mathbf{X}$ is

$$(\mathbf{X}^T\mathbf{X})^{-1} = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}.$$

- Substituting it into the OLS solution gives

$$\begin{aligned}\hat{\Theta} &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} \\ &= \begin{pmatrix} 1 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}.\end{aligned}$$

- Therefore,

$$\hat{\theta}_0 = 3, \quad \hat{\theta}_1 = -1, \quad \hat{\theta}_2 = -2.$$

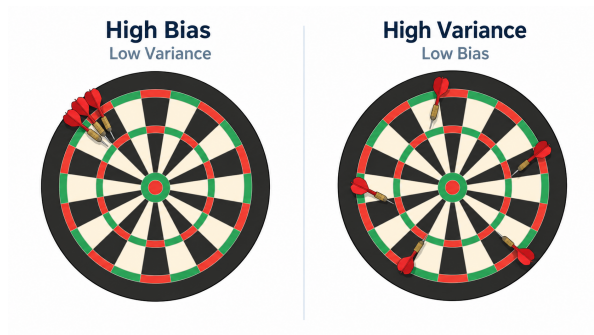
- The fitted linear regression model is

$$\hat{y} = 3 - x_1 - 2x_2.$$



Part 3. Bias-variance decomposition

Bias and variance: intuition



- **Bias** measures how far the average prediction is from the true value.
- **Variance** measures how much the predictions vary across different training datasets.
- High bias means the model is systematically inaccurate.
- High variance means the model is sensitive to the training data.

Problem settings

- Features: X . True relationship: $f(X)$.
- Observed labels: $y = f(X) + \varepsilon$, where ε is a noise, with $E(\varepsilon) = 0$.
- $\hat{f}(X)$ is the estimate of $f(X)$ by some estimators, e.g., OLS.
- Bias-variance decomposition:

$$E[(y - \hat{f})^2] = \text{Bias}[\hat{f}]^2 + \sigma^2 + \text{Var}[\hat{f}]$$

which says the expectation of errors between observed labels and estimated labels is sum of squared bias, squared irreducible errors, and the variance of estimated relationship.

Derivation

- Recall that

$$y = f + \varepsilon, \quad \mathbb{E}[\varepsilon] = 0.$$

- Add and subtract $\mathbb{E}[\hat{f}]$ to decompose the prediction error:

$$\begin{aligned} y - \hat{f} &= f + \varepsilon - \hat{f} \\ &= \underbrace{f - \mathbb{E}[\hat{f}]}_{\text{bias term}} + \underbrace{\mathbb{E}[\hat{f}] - \hat{f}}_{\text{model fluctuation}} + \underbrace{\varepsilon}_{\text{noise}}. \end{aligned}$$

- Taking the squared expectation, the cross terms vanish because

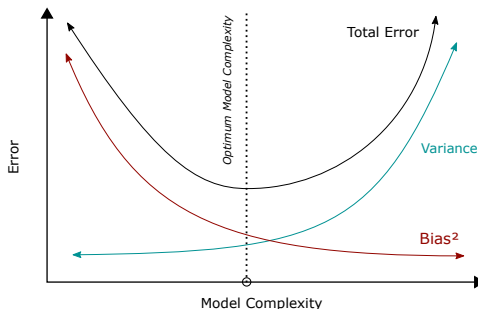
$$\mathbb{E}[\varepsilon] = 0, \quad \mathbb{E}[\mathbb{E}[\hat{f}] - \hat{f}] = 0.$$

- Therefore,

$$\begin{aligned} \mathbb{E}[(y - \hat{f})^2] &= (f - \mathbb{E}[\hat{f}])^2 + \mathbb{E}[(\mathbb{E}[\hat{f}] - \hat{f})^2] + \mathbb{E}[\varepsilon^2] \\ &= \text{Bias}[\hat{f}]^2 + \text{Var}[\hat{f}] + \sigma^2. \end{aligned}$$

Bias-variance tradeoff

- As model complexity increases, bias typically decreases while variance increases.



- High bias is mainly caused by insufficient model capacity.
- High variance is mainly caused by excessive sensitivity to the training data.